actual Software Image data for BDX Protocol clients, when a Software Image is determined to be available from the public Internet or from local storage.

Upon receipt of a QueryImageResponse command containing a DelayedActionTime field, the OTA Requestor SHALL wait for at least the stated delay time before initiating the first part of a file transfer for the URI provided.

The OTA Provider MAY expunge any previously cached Software Image downloaded on behalf of other OTA Requestors, to save storage, at any time, as long as no transfer is currently in active progress. It is RECOMMENDED that OTA Providers on a given Fabric maintain as much Software Image cache as practical, to improve availability of software image and reduce latency between QueryImage requests and availability of the matching QueryImageResponse for a new Software Image ready to be downloaded.

In order to support non-BDX protocols relying on the public Internet, or intranets, the OTA Requestor SHALL only report support for protocols requiring public Internet access if it has determined that it does indeed have access to the necessary network domains beyond the Fabric. The OTA Requestor MAY employ any method it deems satisfactory to determine public Internet reachability. Because of the variety of firewall and security policies on network infrastructure, it is RECOMMENDED that all Nodes whose primary networking interactions lie within protocols in-band of this wider specification support the BDX method, so that even if a Node cannot access the public Internet, it MAY still obtain OTA Software Images by relying on a local OTA Provider which can.

For BDX transfers, the following BDX-specific constraints SHALL be followed:

- Receiver-Drive mode SHALL be used by the OTA Provider for any transfers initiated from a secure channel on non-TCP transport.
- Asynchronous mode MAY be used by the OTA Provider for any transfer initiated from a secure channel on TCP transport. Note that Asynchronous mode is provisional.
- Idle time-out SHALL be no less than 5 minutes for either Receiver-Driver or Asynchronous mode, before aborting a transfer.
- Block sizes constraints SHALL be as follows:
  - Maximum Block Size over all transports SHALL be a power of two if the OTA Requestor requests a value larger than 128 bytes.
  - For an OTA Requestor-requested Maximum Block Size value between 16 and 128, the exact requested value SHALL be used. This constraint allows low-power Nodes to precisely control the block sizes to ensure their power constraints are respected, including enabling single-frame block transfers over communication mediums where MTU is very small.
  - Maximum Block Size requested by OTA Requestors over non-TCP transports SHALL be no larger than 1024 (2^10) bytes. OTA Providers SHALL support the Maximum Block Size of at least 1024 bytes in those cases.
  - Maximum Block Size requested by OTA Requestors over TCP transport SHALL be no larger than 8192 (2 ^ 13) bytes. OTA Providers SHALL support a Maximum Block Size of at least 4096 (2^12) bytes in this case and MAY support 8192 bytes.
  - Actual Block Size used over all transports SHALL be the negotiated Maximum Block Size for every block except the last one, which may be of any size less or equal to the Maximum

